

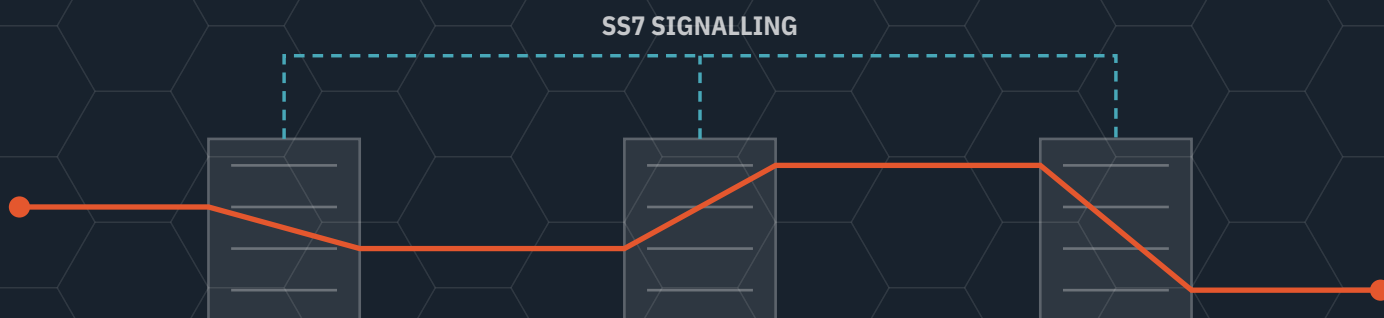
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 2

2G Circuit Switching

سوئیچینگ مداري 2G



Lecture Notes

Instructor: Amin Azari

Compiled by Aidin Shekari

استاد: امین آذری github.com/AminAzari

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1 Circuit Switching Fundamentals

Concept: Dedicated Path

In circuit switching, a **dedicated communication path** is established between two endpoints for the entire duration of the call. This path is reserved exclusively — no other user can use those resources until the call is released.

How It Works in GSM

Step	Action
1	Caller initiates a call
2	Network allocates a dedicated timeslot on the air interface (TDMA)
3	A fixed path is reserved through BTS → BSC → MSC → trunk to destination
4	Resources remain allocated for the entire call duration
5	Upon hangup, all resources are released

Resource Reservation

- **Air Interface:** A dedicated TDMA timeslot (one of 8 per carrier frequency) is assigned
- **Abis Interface** (BTS↔BSC): A 16 kbps or 64 kbps channel on E1/T1 links
- **A Interface** (BSC↔MSC): A 64 kbps timeslot on E1/T1 links
- **Trunk** (MSC↔PSTN/other MSC): A 64 kbps circuit on PCM links

Key Characteristics

- **Connection-oriented:** Path established before data flows
- **Guaranteed bandwidth:** Fixed 64 kbps per direction once connected
- **Constant delay:** Predictable latency (~150ms end-to-end target)
- **Inefficient for bursty data:** Resources wasted during silence periods

2 SS7 Signaling Concepts

What is SS7?

Signaling System 7 (SS7) is the out-of-band signaling protocol stack used by telecom networks to:

- Set up and tear down calls
- Manage subscriber databases
- Enable roaming and handovers
- Provide supplementary services (call forwarding, SMS, etc.)

SS7 signaling is **separate** from the voice path — this is called **Common Channel Signaling (CCS)**.

SS7 Protocol Stack (Simplified)

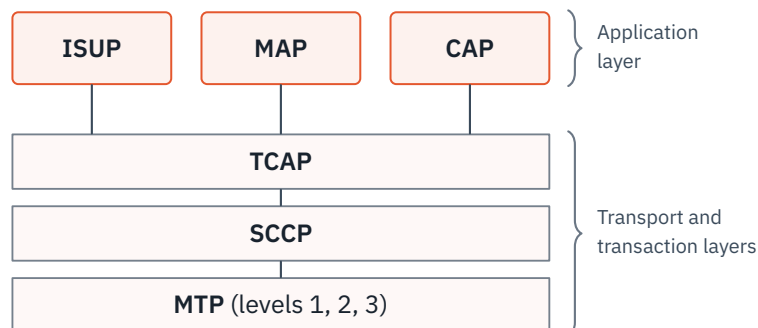


Figure 2.1 Simplified SS7 protocol stack

Key Protocols

ISUP (ISDN User Part)

- **Purpose:** Call setup, management, and release between exchanges (MSCs)
- **Messages:** IAM (Initial Address Message), ACM (Address Complete), ANM (Answer), REL (Release), RLC (Release Complete)
- **Used for:** Establishing the voice circuit between switches

■ MAP (Mobile Application Part)

- **Purpose:** Communication between GSM network elements (MSC, HLR, VLR, AuC)
- **Operations:** Location updates, authentication data retrieval, subscriber info queries, SMS routing
- **Used for:** Mobility management, roaming, subscriber data exchange

■ TCAP (Transaction Capabilities Application Part)

- **Purpose:** Provides a framework for MAP and other applications to exchange structured data
- **Model:** Request-response transactions (invoke, result, error)
- **Used for:** Carrying MAP operations over the SS7 network

■ Signaling Points

Element	Role
SSP (Service Switching Point)	MSC/GMSC — originates/terminates signaling
STP (Signal Transfer Point)	Routes signaling messages between SSPs
SCP (Service Control Point)	Database nodes (HLR, AuC)

3 Call Setup Process

3.1 Mobile-Originated Call (MOC)

A Mobile-Originated Call is initiated by the mobile subscriber.

■ Sequence of Events

1. **UE dials number** → sends CHANNEL REQUEST on RACH
2. **BTS** forwards to BSC
3. **BSC** assigns a dedicated channel (SDCCH) → IMMEDIATE ASSIGNMENT
4. **UE** sends CM SERVICE REQUEST (service type = MOC) to MSC via BSC
5. **MSC** authenticates the subscriber (see Section 6)
6. **MSC** activates ciphering
7. **UE** sends SETUP message with called party number
8. **MSC** sends CALL PROCEEDING to UE
9. **MSC** analyzes digits, determines routing
10. **MSC** sends IAM (ISUP) toward GMSC/PSTN
11. **GMSC/PSTN** returns ACM (Address Complete) → ringback
12. **MSC** sends ALERTING to UE
13. **Called party answers** → ANM (Answer Message) returned
14. **MSC** sends CONNECT to UE

15. **BSC** assigns a TCH (Traffic Channel) for voice
16. **Voice path established** — conversation begins

■ MOC Sequence Diagram

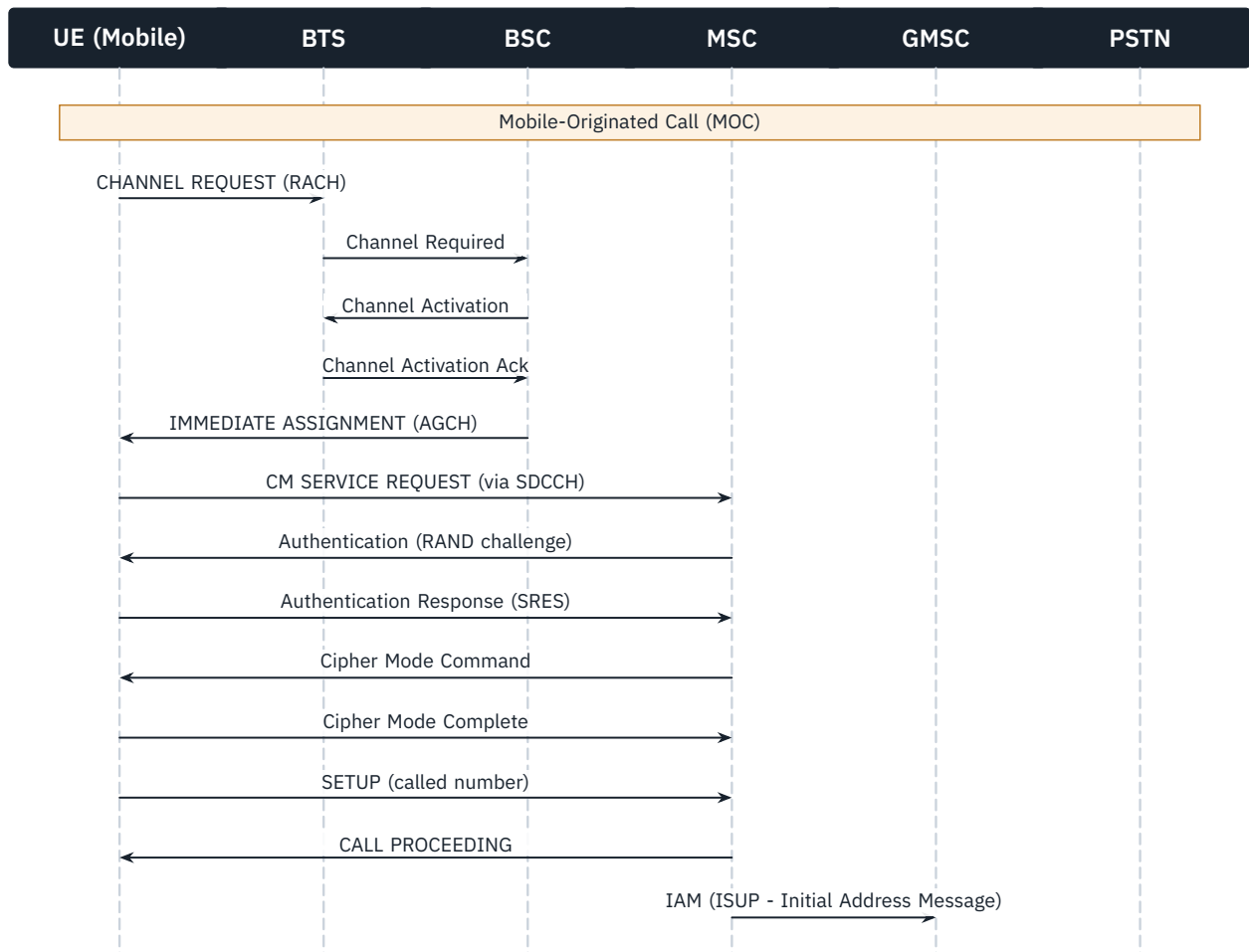


Figure 2.2 Mobile-originated call (MOC) set-up (part 1 of 2)

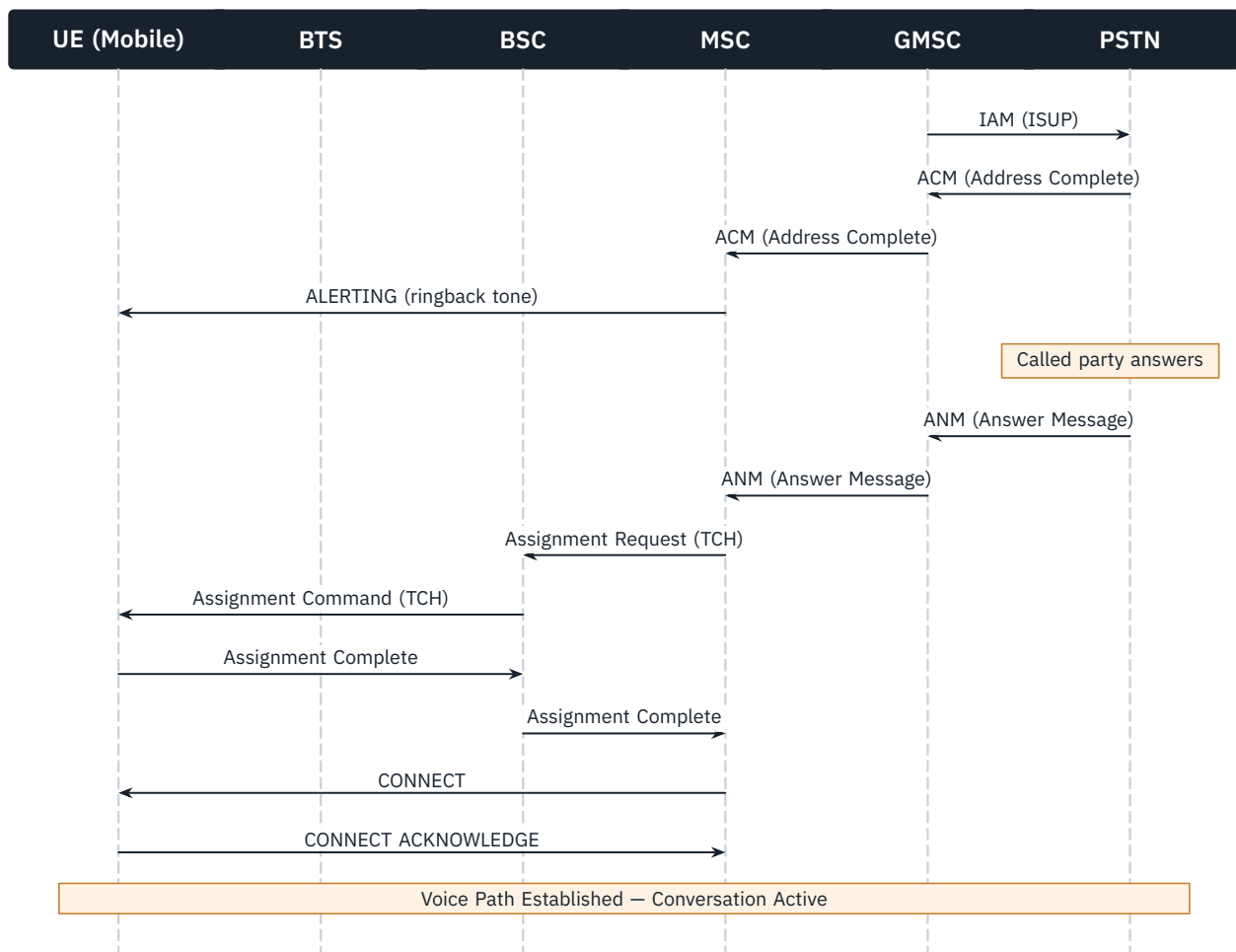


Figure 2.3 Mobile-originated call (MOC) set-up (part 2 of 2)

3.2 Mobile-Terminated Call (MTC)

A Mobile-Terminated Call is a call arriving for a mobile subscriber.

■ Sequence of Events

1. **PSTN/calling party** sends IAM to GMSC with called MSISDN
2. **GMSC** queries HLR: “Where is this subscriber?” (MAP: Send Routing Info)
3. **HLR** returns MSRN (Mobile Station Roaming Number) pointing to serving MSC
4. **GMSC** routes call to serving MSC using MSRN (IAM)
5. **MSC** queries VLR for subscriber state and location area
6. **MSC** pages the UE in the Location Area
7. **UE** responds to paging on RACH
8. **BSC** assigns SDCCH → authentication & ciphering
9. **MSC** sends SETUP to UE
10. **UE** sends CALL CONFIRMED
11. **UE** alerts user (ringtone)
12. **User answers** → UE sends CONNECT
13. **MSC** sends ANM back toward GMSC/PSTN
14. **Traffic channel assigned** — voice path active

■ MTC Sequence Diagram

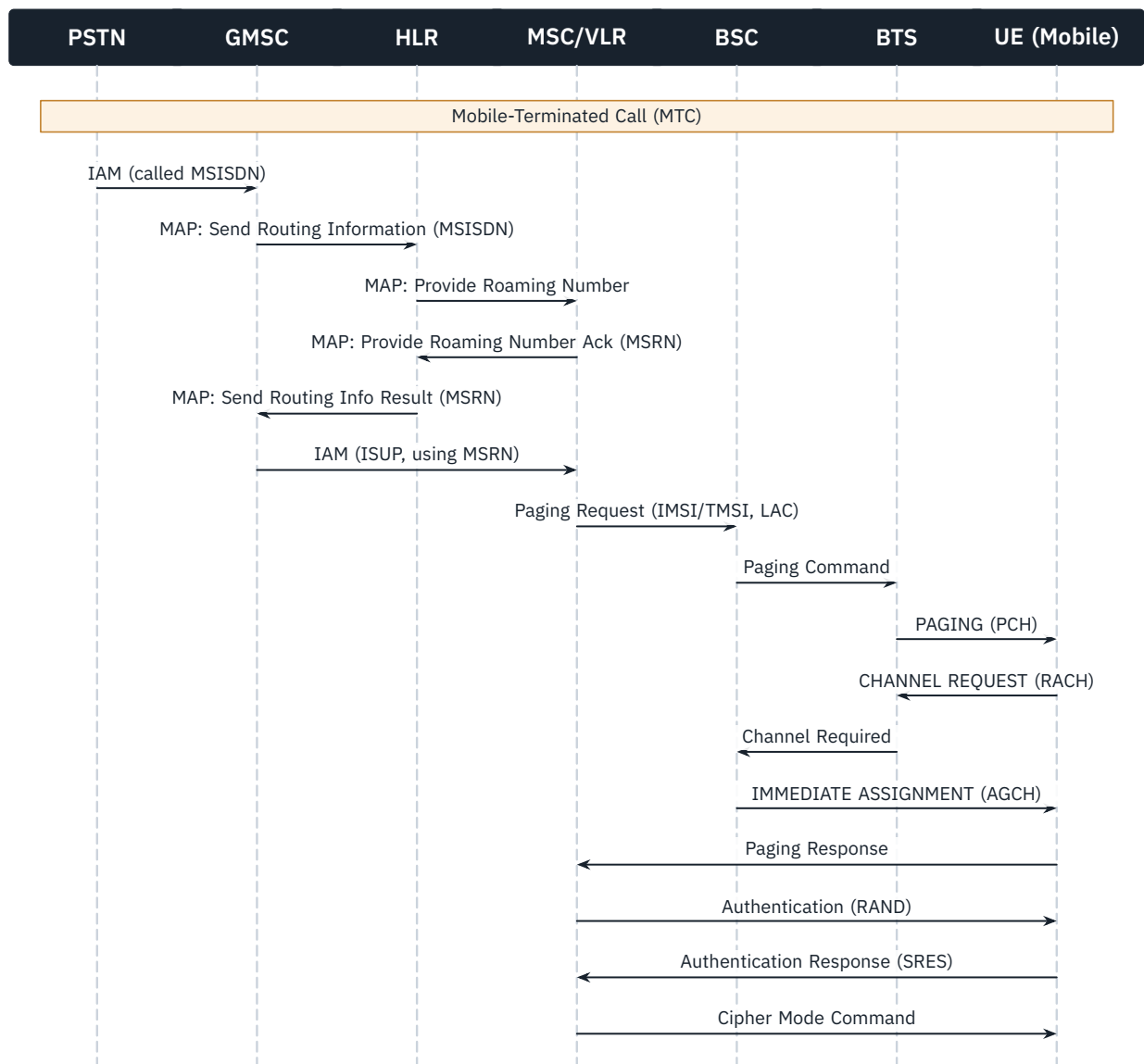


Figure 2.4 Mobile-terminated call (MTC) set-up (part 1 of 2)

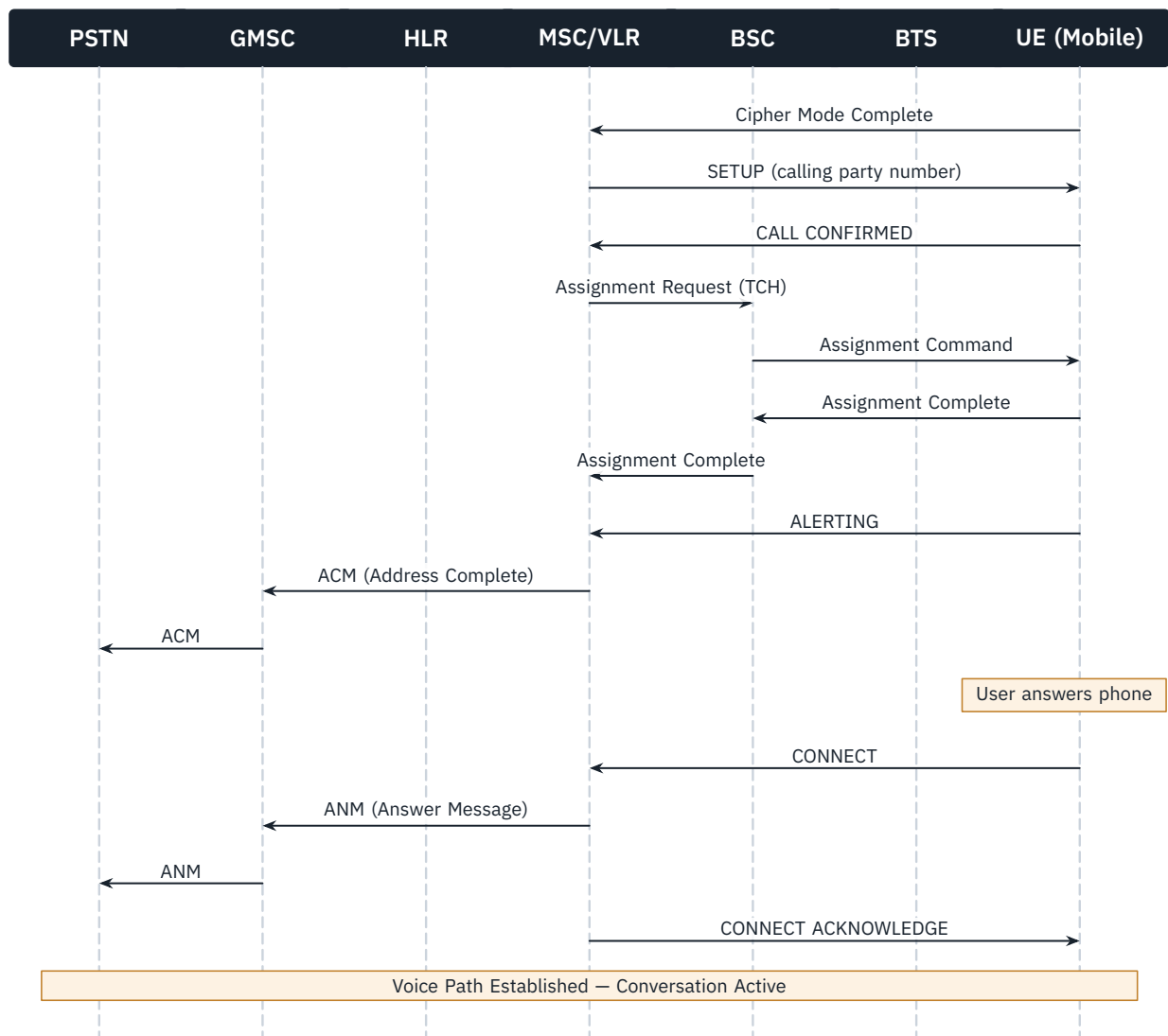


Figure 2.5 Mobile-terminated call (MTC) set-up (part 2 of 2)

4 Call Routing

How MSC/GMSC Route Calls Using HLR

The HLR is the **master database** of all subscribers. It knows which MSC/VLR currently serves each subscriber.

Routing a Mobile-Terminated Call

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Step 1: PSTN delivers call to GMSC (based on MSISDN number plan)
Step 2: GMSC sends MAP "Send Routing Information" to HLR
Step 3: HLR identifies the serving MSC/VLR for that subscriber
Step 4: HLR asks serving VLR to allocate an MSRN (temporary routing number)
Step 5: VLR returns MSRN to HLR
Step 6: HLR returns MSRN to GMSC
Step 7: GMSC uses MSRN to route the ISUP IAM to the correct MSC
    
```

MSRN (Mobile Station Roaming Number)

- Temporary number in E.164 format
- Looks like a regular phone number in the serving network's numbering plan
- Allows GMSC to route the call using standard ISUP without knowing GSM internals
- Allocated per call, released after routing

Routing a Mobile-Originated Call

For MOC, routing is simpler:

1. MSC receives the dialed digits from the UE
2. MSC performs digit analysis (number plan, prefixes, supplementary services)
3. MSC routes the call via ISUP toward the appropriate trunk/GMSC/PSTN gateway
4. If calling another mobile on same network → MSC may query HLR for MSRN

5 Location Update Procedure

Why Location Updates Are Needed

The network must know **where a subscriber is** to deliver calls and messages. Without location tracking:

- The network would have to page every cell in the entire country
- This would overwhelm the paging channel capacity
- Call setup time would be unacceptable

Location updates allow the network to **narrow paging to a specific Location Area** (group of cells).

Types of Location Updates

Type	Trigger
IMSI Attach	Phone powers on — informs network subscriber is reachable
Normal LU	UE moves to a new Location Area (detects new LAI on BCCH)
Periodic LU	Timer-based update to confirm UE is still reachable
IMSI Detach	Phone powers off gracefully — informs network

Location Update Sequence Diagram

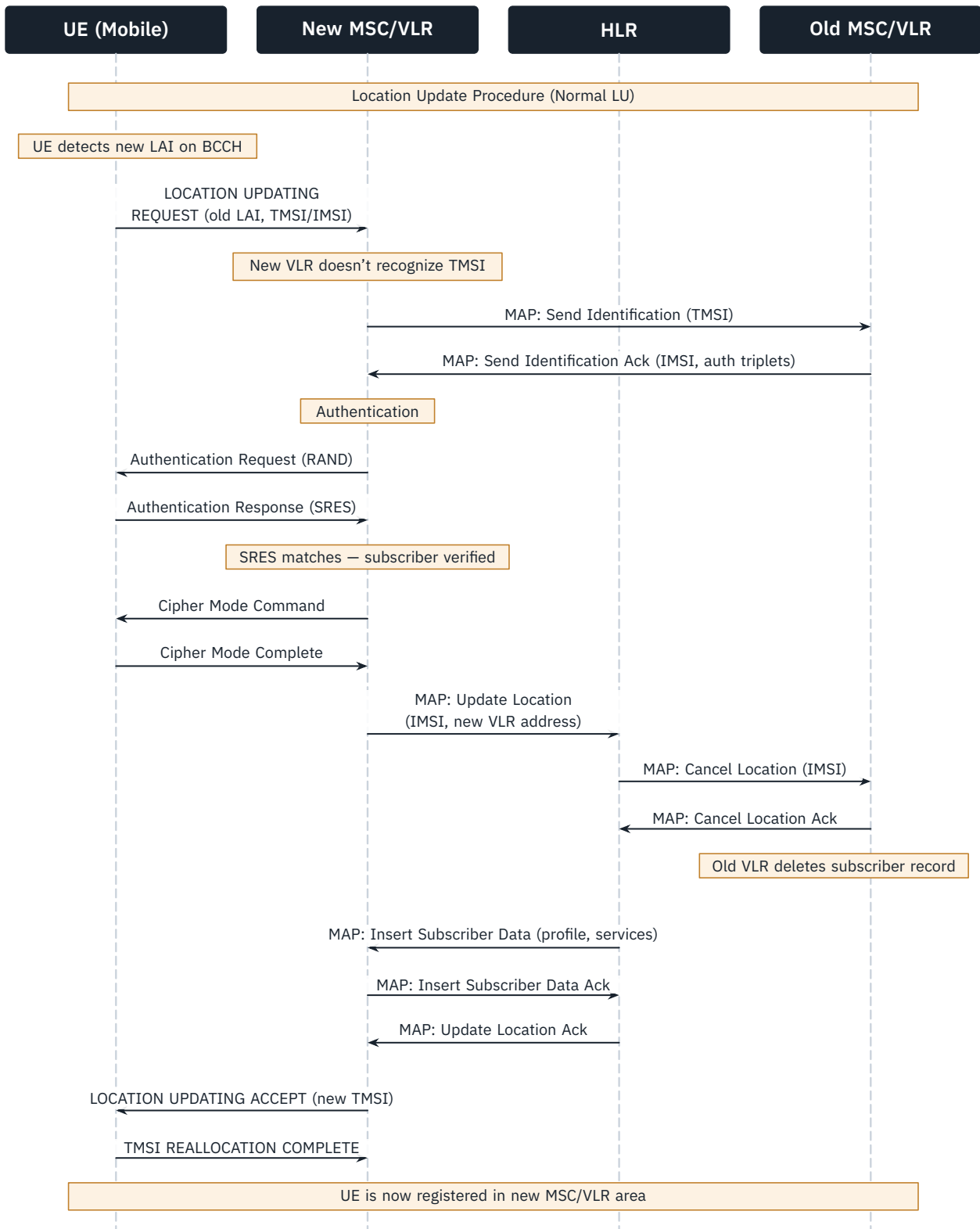


Figure 2.6 Location update with a change of MSC/VLR

IMSI Attach

When a phone powers on:

1. UE reads BCCH to get LAI and network information
2. UE sends LOCATION UPDATING REQUEST with attach flag
3. Same procedure as normal LU follows
4. VLR marks subscriber as “attached” (reachable)

Periodic Location Update

- Network broadcasts a timer value (T3212) on BCCH
- UE must send a LU before the timer expires, even if it hasn’t moved
- If the timer expires without an LU, VLR marks subscriber as “implicitly detached”
- Purpose: detect phones that powered off without IMSI Detach (battery died, etc.)

6 Authentication Procedure

Purpose

Authentication ensures that the subscriber is who they claim to be, preventing:

- Unauthorized network access
- Cloned SIM fraud
- Eavesdropping (when combined with ciphering)

The Authentication Triplet

The AuC (Authentication Center, co-located with HLR) generates **authentication triplets**:

Element	Size	Purpose
RAND	128 bits	Random challenge sent to UE
SRES	32 bits	Expected response (computed from RAND + Ki)
Kc	64 bits	Cipher key for encrypting the air interface

How Triplets Are Generated

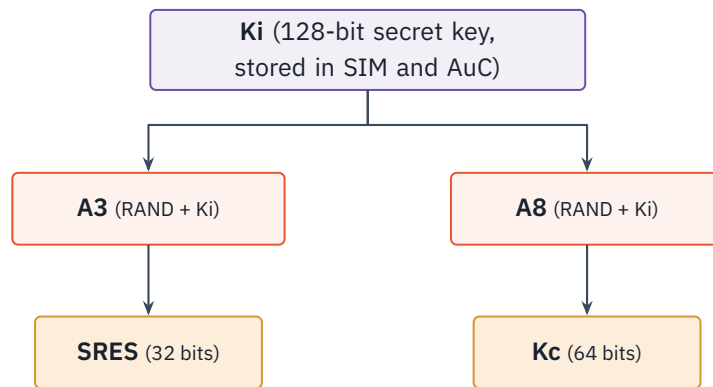


Figure 2.7 Generation of an authentication triplet (RAND, SRES, Kc)

- **Ki** never leaves the SIM card or the AuC
- The same computation happens independently on both sides
- If SRES from UE matches SRES from triplet → subscriber is authenticated

Authentication Sequence Diagram

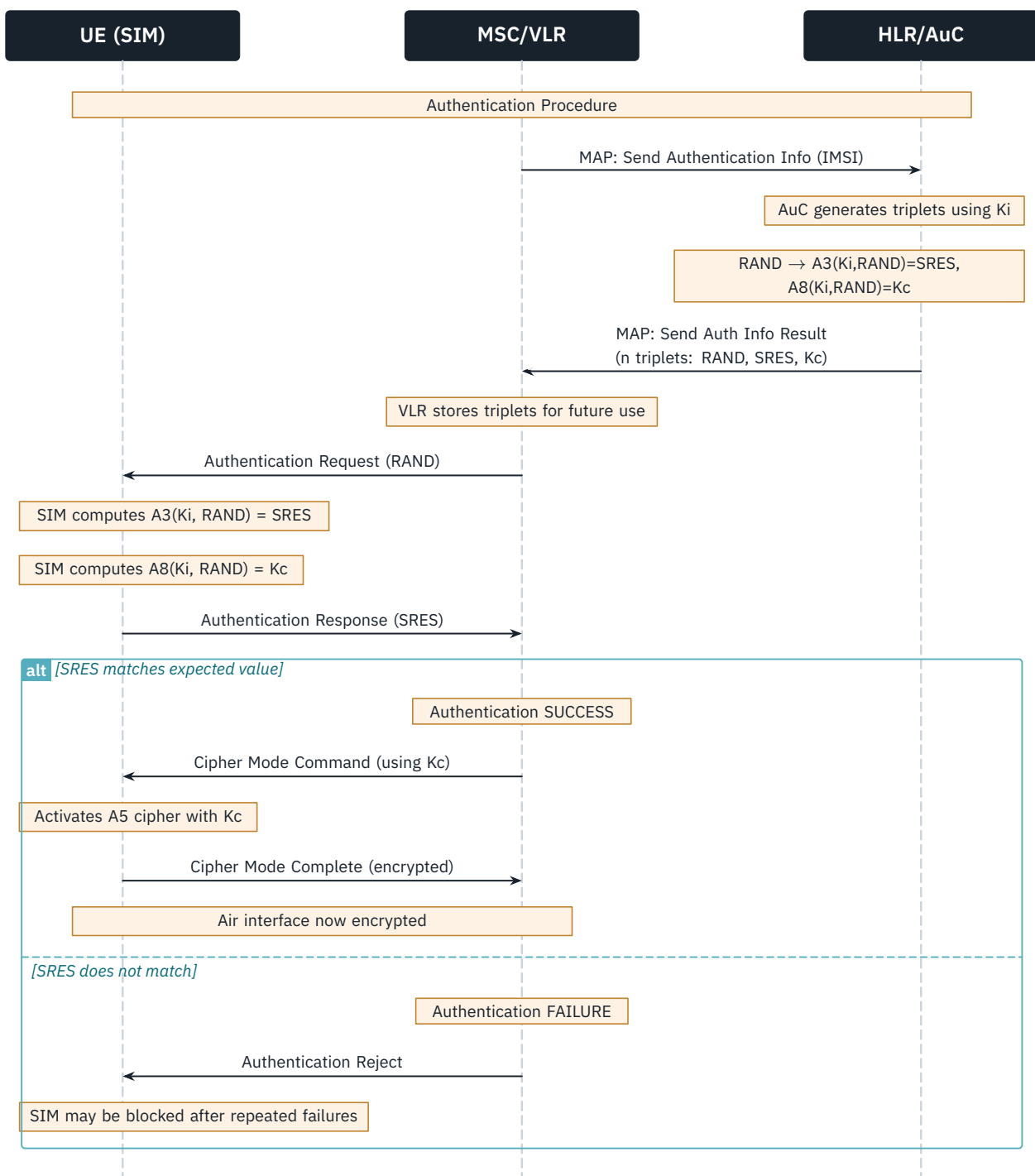


Figure 2.8 GSM authentication and ciphering

Key Points

- VLR stores **multiple triplets** (typically 5) to avoid querying HLR for every transaction
- Authentication is performed at: call setup, location update, SMS, supplementary service activation
- GSM authentication is **one-way** – network authenticates the UE, but UE does not authenticate the network (vulnerability fixed in 3G/UMTS)

7 Roaming

Concepts

Term	Definition
HPLMN	Home PLMN — the network where the subscriber has their subscription
VPLMN	Visited PLMN — a foreign network the subscriber is using while roaming
Roaming Agreement	Commercial + technical agreement between two operators
GRX/IPX	Interconnect network for signaling between PLMNs

How Roaming Works

1. **UE powers on in foreign country** → selects VPLMN (based on SIM's preferred PLMN list)
2. **UE performs Location Update** to VPLMN's MSC/VLR
3. **VPLMN's VLR** contacts **HPLMN's HLR** via SS7/MAP (through international signaling links)
4. **HLR** authenticates subscriber (sends triplets to VPLMN)
5. **HLR** sends subscriber profile to VPLMN's VLR
6. **HLR** records the VPLMN's VLR address as the subscriber's current location
7. **Subscriber can now make/receive calls** through VPLMN

Signaling Path for Roaming

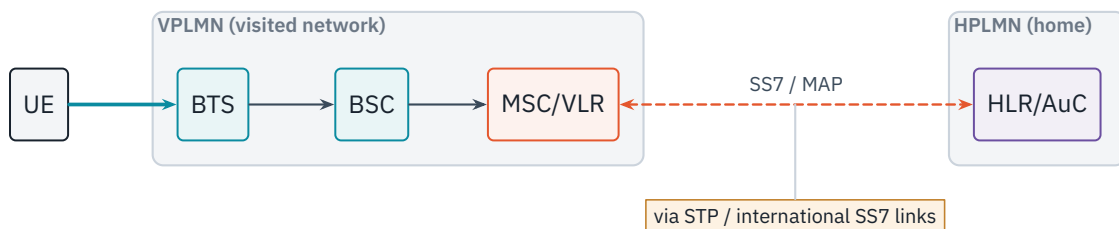


Figure 2.9 Signalling path for roaming: visited MSC/VLR to home HLR/AuC over SS7/MAP

Incoming Call to a Roaming Subscriber

1. Call arrives at HPLMN's GMSC
2. GMSC queries HLR → learns subscriber is in VPLMN
3. HLR requests MSRN from VPLMN's VLR
4. VPLMN's VLR returns MSRN
5. HLR gives MSRN to GMSC
6. GMSC routes call to VPLMN's MSC using MSRN (via international trunks)
7. VPLMN's MSC pages and connects the subscriber

Outgoing Call from Roaming Subscriber

1. UE initiates call in VPLMN
2. VPLMN's MSC handles the call locally
3. Call is routed from VPLMN to destination (may transit through HPLMN or go direct)
4. CAMEL/CAP protocol may be used for prepaid control from HPLMN

8 Call Release

Normal Call Release (Mobile-Initiated)

1. **UE** sends DISCONNECT to MSC
2. **MSC** sends REL (Release) via ISUP toward the far end
3. **Far end** responds with RLC (Release Complete)
4. **MSC** sends RELEASE to UE
5. **UE** sends RELEASE COMPLETE
6. **BSC** deallocates the TCH (traffic channel)
7. **All resources freed**: timeslots, trunks, circuit references

Network-Initiated Release

Can be triggered by:

- Far-end hangup
- Radio link failure (UE out of coverage)
- BSC/MSC timer expiry
- Administrative disconnect

ISUP Release Messages

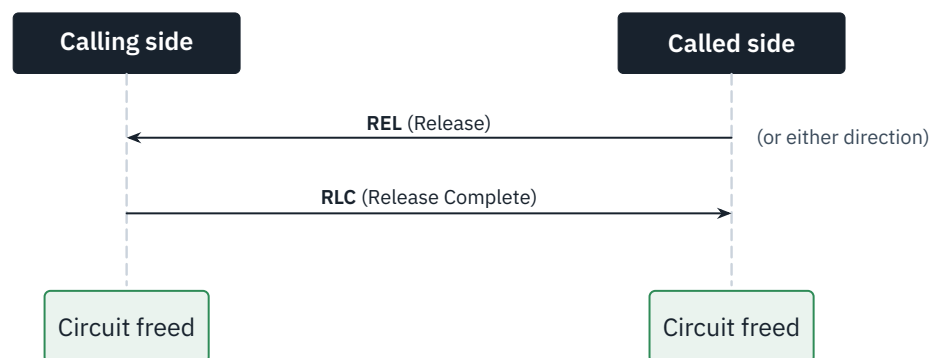


Figure 2.10 ISUP call release

9 Why Circuit Switching?

Why Circuit Switching Was Used for Voice

Voice Requirements

- **Constant bit rate:** 13 kbps (full rate) or 6.5 kbps (half rate) — continuous stream
- **Low and predictable delay:** < 150ms one-way for acceptable conversation quality
- **No jitter tolerance:** Human ear is very sensitive to variable delay
- **Symmetric traffic:** Both parties speak roughly equally

Why Circuit Switching Fits Voice

Requirement	How CS Addresses It
Low delay	Dedicated path = no queuing delay
No jitter	Fixed timeslot allocation = constant delay
Guaranteed bandwidth	Reserved resources = no congestion drops
Simple processing	Once path is set up, switching is trivial
Real-time	No store-and-forward delays

Historical Context (1980s-90s)

- Processing power was expensive — simple switching was preferred
- Memory was expensive — no buffering/packetization
- Existing PSTN was circuit-switched — natural integration
- Voice was the **only** service expected from mobile phones

Why Circuit Switching Became a Limitation

The Problem with Data

Issue	Explanation
Resource waste	A 64 kbps circuit is held even during silence (50-60% of a voice call)
No statistical multiplexing	Cannot share resources among multiple users dynamically
Fixed bandwidth	Cannot burst above 64 kbps even when network is idle
Call setup delay	3-7 seconds to establish a circuit — unacceptable for web browsing
Per-minute charging	Inefficient for checking email (30 seconds of data, charged 1 minute)
Scalability	Each data session consumes a full voice circuit

Data Traffic Characteristics vs. Voice



Figure 2.11 Occupancy of a reserved circuit: continuous voice versus bursty data

Circuit switching **cannot efficiently handle bursty traffic** because resources are reserved but unused most of the time.

The Evolution Path

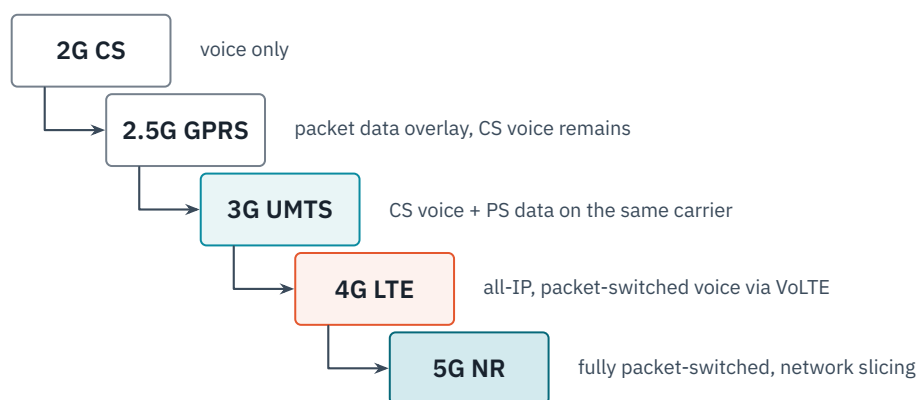


Figure 2.12 From circuit switching to fully packet-switched networks

Summary

Circuit switching was the **right choice** for voice in the 1990s — it provided the quality guarantees that voice communication demands with the technology available at that time. However, as mobile usage shifted from voice-only to data-centric services, the inefficiency of holding dedicated circuits for bursty, asymmetric data traffic made packet switching the inevitable successor.

Key Terms Summary

Term	Definition
MOC	Mobile-Originated Call
MTC	Mobile-Terminated Call
MSRN	Mobile Station Roaming Number — temporary number for call routing
IAM	Initial Address Message — ISUP message to initiate a call
ACM	Address Complete Message — called party is being alerted
ANM	Answer Message — called party has answered
REL/RLC	Release / Release Complete — call teardown
LAI	Location Area Identity — identifies a group of cells
TMSI	Temporary Mobile Subscriber Identity — privacy protection
Ki	Secret key shared between SIM and AuC
SRES	Signed Response — authentication proof
Kc	Cipher key derived during authentication
VPLMN	Visited PLMN — foreign network during roaming
HPLMN	Home PLMN — subscriber's home network

Next Module: Module 3 — GPRS and Packet Switching